QUESTION BANK

Unit 1

1. Define Computer Architecture and Computer Organization.
2. What is the Von Neumann Architecture?
3. List the functional units of a computer system.
4. What do the terms MIPS and FLOPS stand for?
5. Explain the difference between machine language and assembly language.
6. Describe the working principle of the Von Neumann Architecture with a neat block diagram.
7. Explain the concept of system bus in a computer system.
8. Differentiate between throughput and latency as performance metrics.
9. Convert the decimal number (–25) into binary using 2's complement representation.
10. Convert the hexadecimal number 3A into its binary equivalent.
11. Given a floating-point number, explain how it is represented using the IEEE 754 standard.
12. Compare SISD, SIMD, MISD, and MIMD architectures with suitable examples.
13. Analyze the role of bus interconnection in improving system performance.
14. Evaluate the impact of floating-point representation errors on computational accuracy.
15. Design a simple block diagram of a computer system showing all functional units and explain their interaction.

Unit 2

1. Define Central Processing Unit (CPU) and state its main components.
2. What is the function of the Arithmetic Logic Unit (ALU)?
3. List the types of instructions used in a CPU.
4. What is meant by an interrupt in a computer system?
5. Explain the structure and functions of the CPU with a neat diagram.
6. Describe the instruction cycle and execution process.
7. Explain the difference between hardwired control unit and microprogrammed control unit.
8. Explain the role of program counter (PC) and stack pointer (SP) in program execution.
9. Show how an instruction is executed using different addressing modes with examples.
10. Illustrate the use of general-purpose and special-purpose registers in instruction execution.
11. Demonstrate how the CPU handles an interrupt during program execution.
12. Analyze the instruction formats and their impact on CPU performance.
13. Compare instruction pipelining and arithmetic pipelining.
14. Evaluate the effectiveness of pipelining in improving CPU performance, mentioning its limitations.

15. Design a simple CPU block diagram showing ALU, CU, registers, and data paths, and explain their interaction.

Unit 3

1. Define memory hierarchy and list its levels.
2. What is the function of cache memory?
3. List different cache mapping techniques.
4. What is meant by virtual memory?
5. Differentiate between RAM and ROM with respect to characteristics and types.
6. Explain paging as a virtual memory management technique.
7. Describe the role of MMU in memory management.
8. Explain the concept of cache coherence in multiprocessor systems.
9. Illustrate direct mapping technique with a suitable example.
10. Explain modern memory technologies: DRAM, SRAM, and Flash memory, highlighting their characteristics, advantages, and applications.
11. Demonstrate the use of TLB in address translation.
12. Compare direct, associative, and set-associative cache mapping techniques.
13. Analyze the causes and effects of memory fragmentation**.**
14. Evaluate different RAID levels in terms of performance, reliability, and cost.
15. Design a memory hierarchy diagram and explain how it optimizes system performance.

Unit 4

1. Define an I/O system.
2. What is Direct Memory Access (DMA)?
3. List the types of I/O devices.
4. What is meant by memory-mapped I/O?
5. Explain the difference between memory-mapped I/O and isolated I/O.
6. Describe interrupt-driven I/O communication technique.
7. Explain the working of DMA with a neat diagram.
8. Describe the role of I/O interfaces such as USB and PCI.
9. Illustrate the programmed I/O data transfer technique with an example.
10. Demonstrate how interrupt processing occurs during I/O operations.
11. Show how buffering improves I/O performance.
12. Compare polling, interrupt-driven I/O, and DMA in terms of CPU involvement and performance.
13. Analyze the need for synchronization techniques like semaphores and spinlocks in I/O systems.
14. Evaluate different I/O performance improvement techniques such as spooling, buffering, and caching.
15. Design a block diagram of an I/O subsystem showing CPU, memory, I/O controller, and devices, and explain the data flow.